



HEXAGON

Multiphysics

Enabling Holistic Design

Melinda Corley

2023

Simulate All Physics in One Place



STRUCTURES



FLUID DYNAMICS



ACOUSTICS & NVH



Durability and
Robustness



CO-SIMULATION



HxGN Multiphysics

Hexagon's **Multiphysics suite** uses cutting-edge, physics-based simulation technologies to solve real-world engineering problems.



MSC
Apex



Patran



MSC
Nastran



Marc



CAE
Fatigue



MSC
CoSim



Dytran



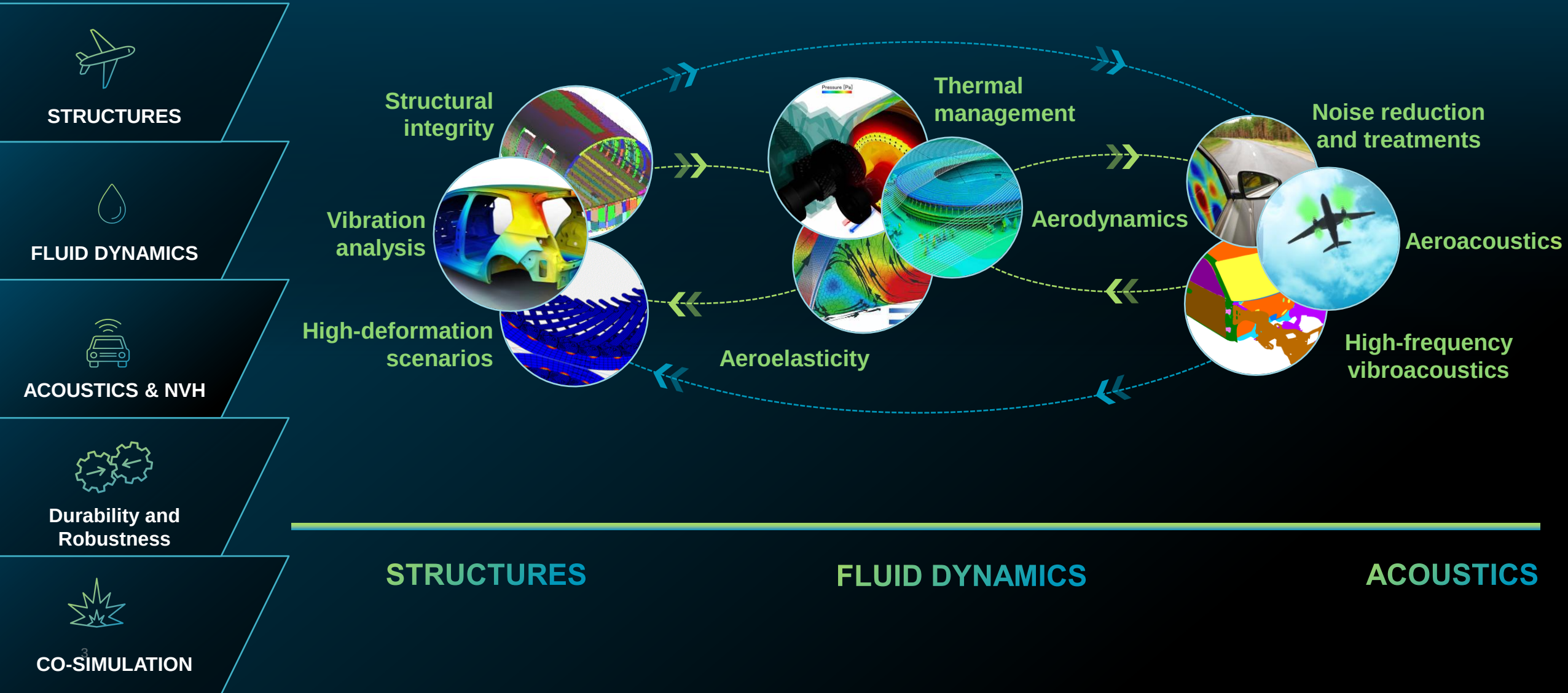
Actran



Cradle
CFD



Circular Designs | Connect, Co-simulate, and Co-design



Indoor Air Circulation with Cradle CFD & MSC Nastran Co-Simulation



STRUCTURES



FLUID DYNAMICS



ACOUSTICS & NVH

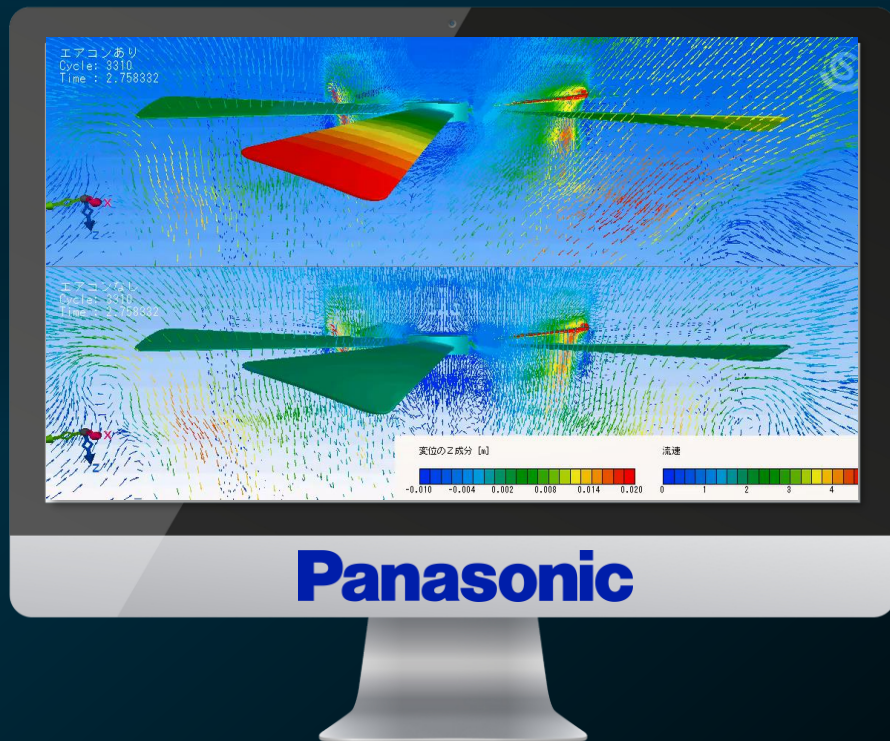


Durability and
Robustness



CO-SIMULATION

hexagonmi.com



Cradle CFD has played a vital role in the development process. We can observe complicated behavior by looking at simulation results in animation format.

Mr. Masahiro Shigemori, Chief of Ventilation Technology Development Division at Panasonic

[Read the article](#)



Panasonic



STRUCTURES



FLUID DYNAMICS



ACOUSTICS & NVH



Durability and
Robustness



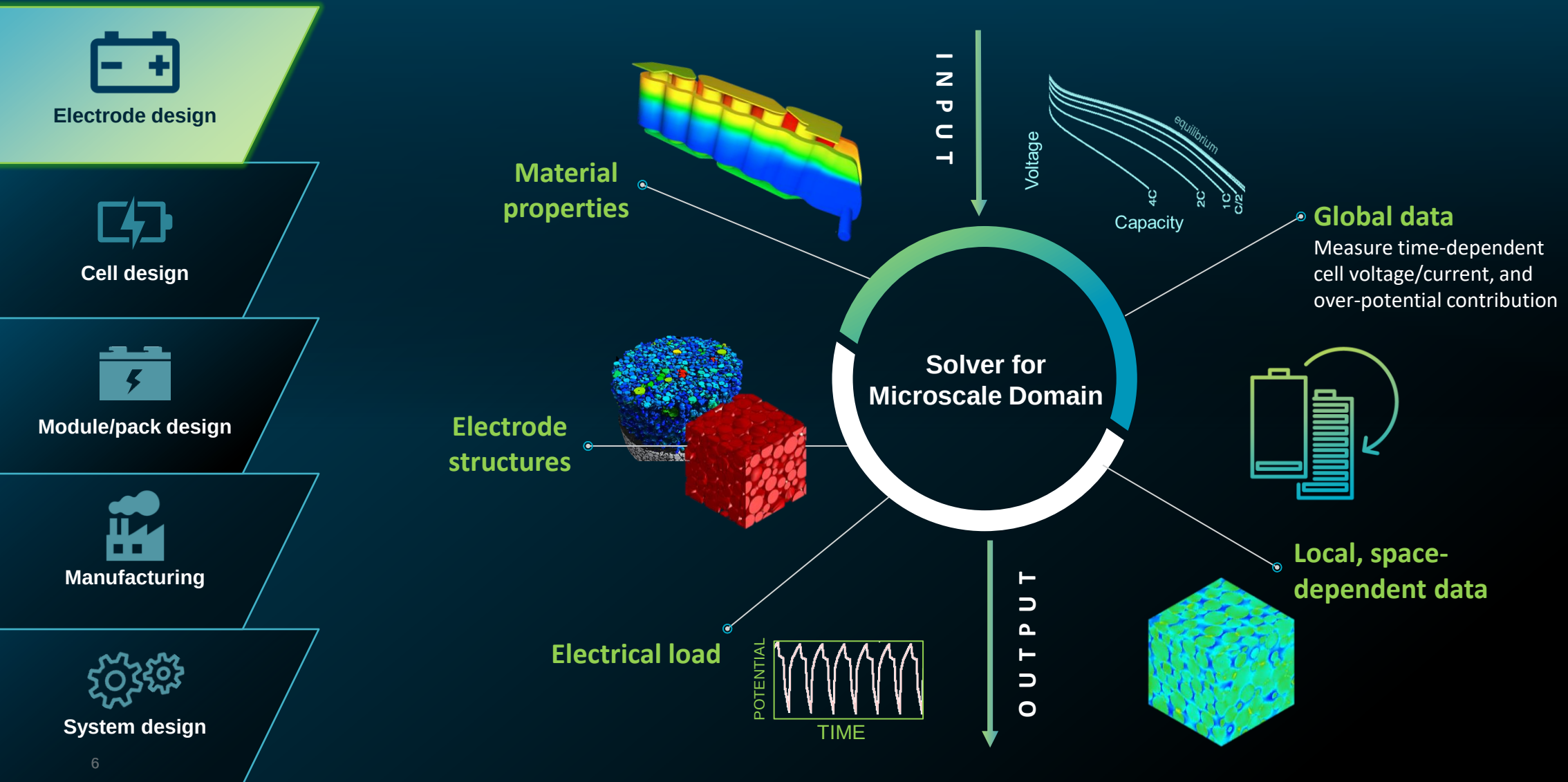
CO-SIMULATION

A holistic approach to batteries

- Gain comprehensive control over **all stages** of development
- Simplify adoption and **compatibility** challenges
- Address industry-specific **requirements**
- Offer more than **individual components**
- **Integrate** electrochemical, mechanical, thermal, and control systems
- Reduce **integration complexity**



Electrode design | From electrochemistry to battery life inputs

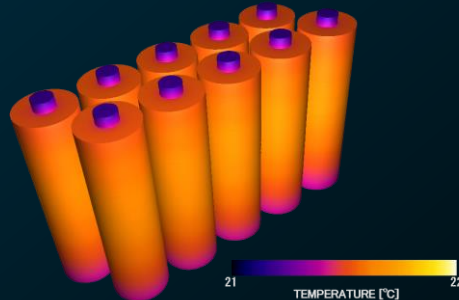
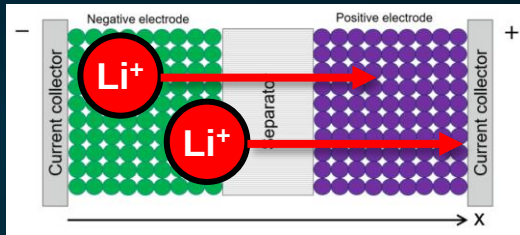


Cell design | Unit level performance for battery life



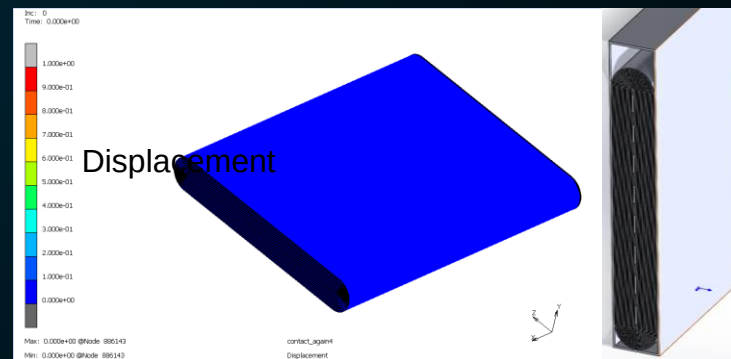
Electrochemical Model

$$\begin{aligned}\nabla \cdot \vec{i}_s &= \nabla \cdot (-\sigma \phi_s) = j^{\text{Li}} \\ \nabla \cdot (k_D^{\text{eff}} \nabla \phi_e) + \nabla \cdot (k_D^{\text{eff}} \nabla \ln c_e) &= -j^{\text{Li}} \\ j^{\text{Li}} &= a_s i_0 \left[\exp \left(\frac{a_a F \eta}{RT} \right) - \exp \left(-\frac{a_c F \eta}{RT} \right) \right] \\ \frac{\partial c_s}{\partial t} &= \nabla \cdot (D_s \nabla c_s) \\ \frac{\partial (\epsilon_e c_e)}{\partial t} &= \nabla \cdot (D_e^{\text{eff}} \nabla c_e) + \frac{1 - t_+^0}{F} j^{\text{Li}}\end{aligned}$$

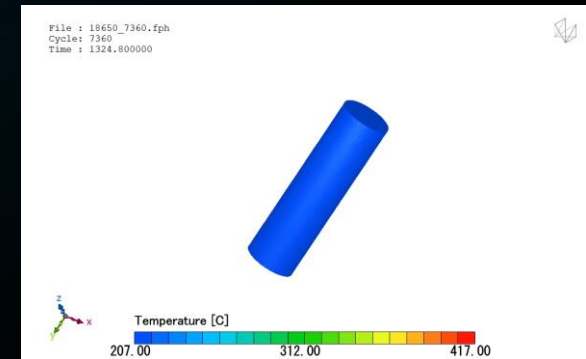


- ✓ Accurate prediction of cell behavior using physics-based modeling
- ✓ Phenomenological approach for cell swelling
- ✓ Light-weight exothermic reaction model for fast prediction of cell thermal runaway

Cell swelling



Thermal Runaway



Module & Pack Design | Virtual Validation for real-world scenarios

Thermal & Electro-thermal Simulation

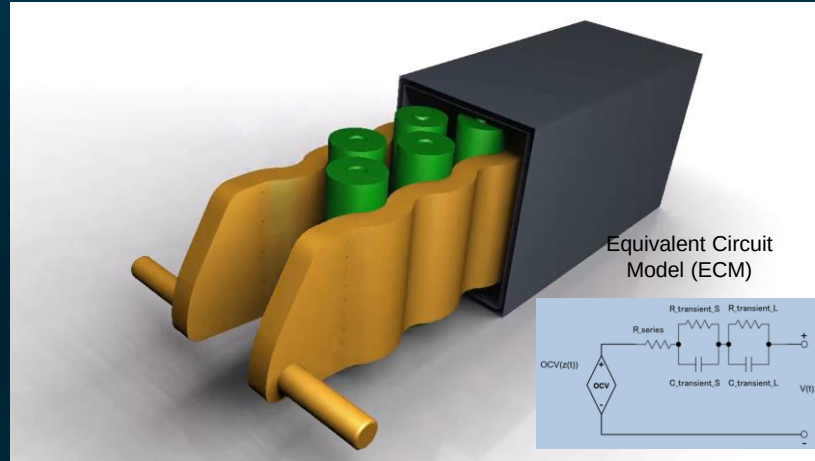

Electrode design


Cell design

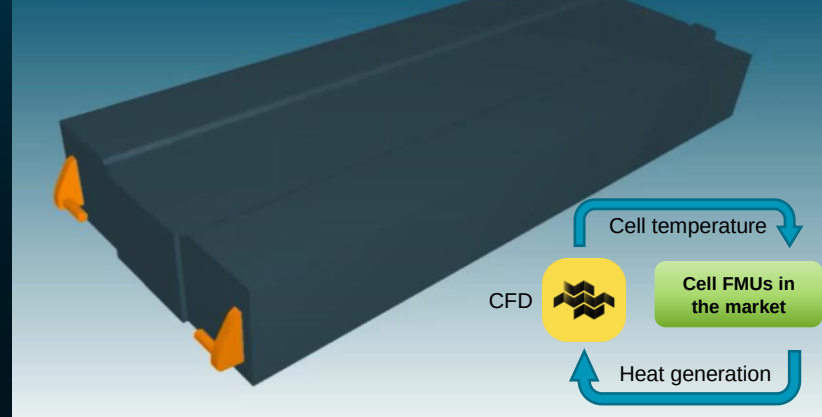

Module/pack design


Manufacturing


System design



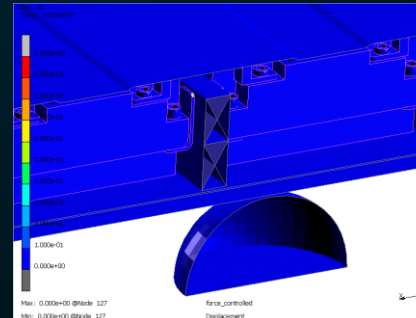
Functional Mock-up Interface (FMI) for external cell models



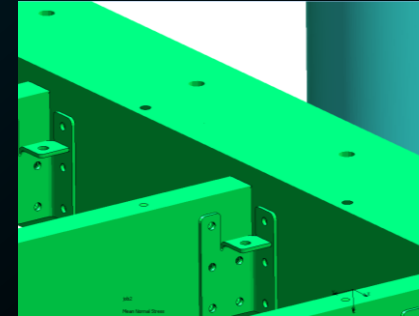
Mechanical Simulation



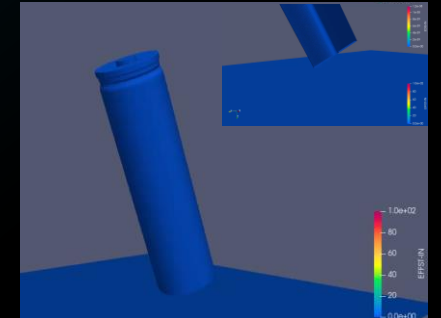
Fatigue analysis



Intrusion



Crush



Drop

Manufacturing | Improving yield, building trust in the design


Electrode design

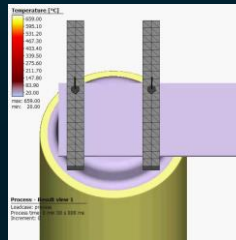
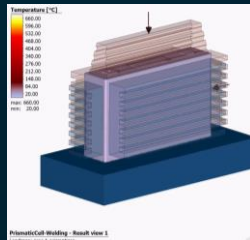
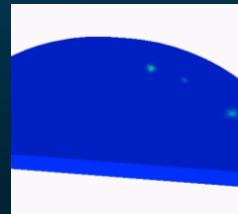
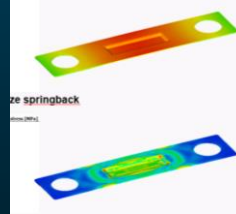
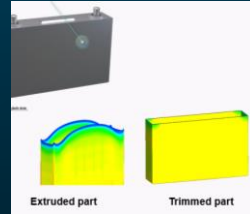

Cell design


Module/pack design


Manufacturing

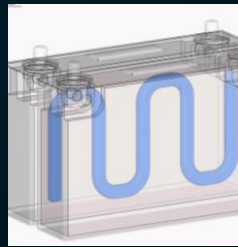
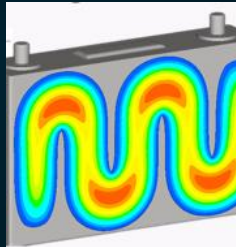
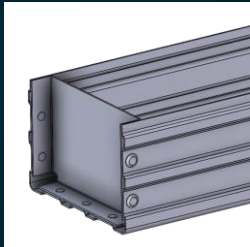

System design

CELLS



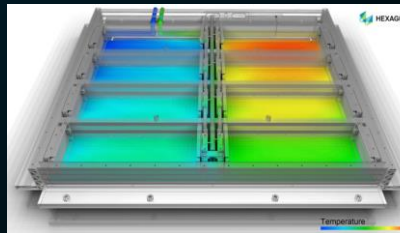
- ✓ Forming simulation
- ✓ Riveting / welding simulation

MODULE



- ✓ Riveting / bolting simulation
- ✓ Flow analysis of adhesives for modules assembly

TRAY



- ✓ Component design
- ✓ Forming process engineering
- ✓ Assembly and joining process

System Design | From battery pack to entire vehicle system



Electrode design



Cell design



Module/pack design

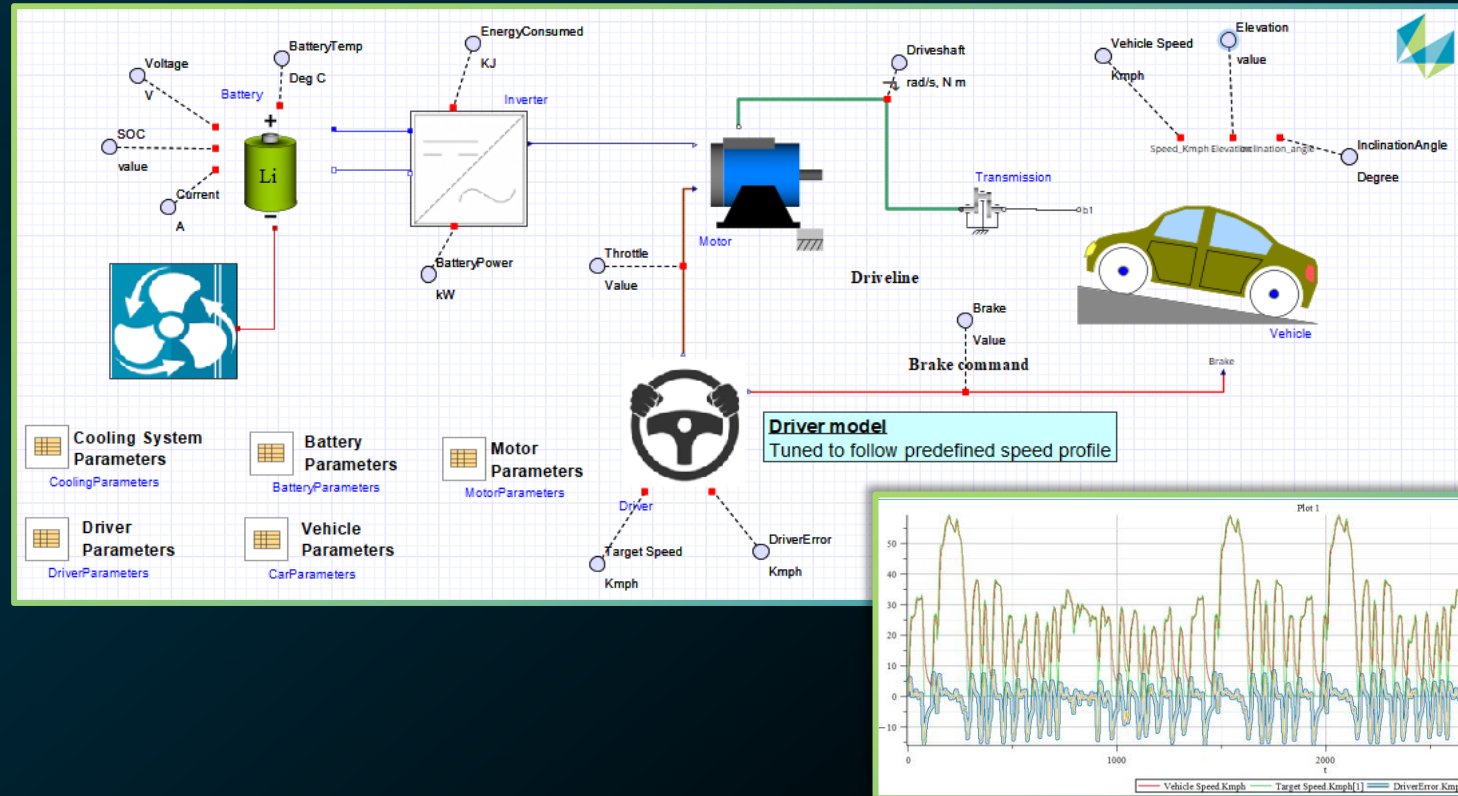


Manufacturing

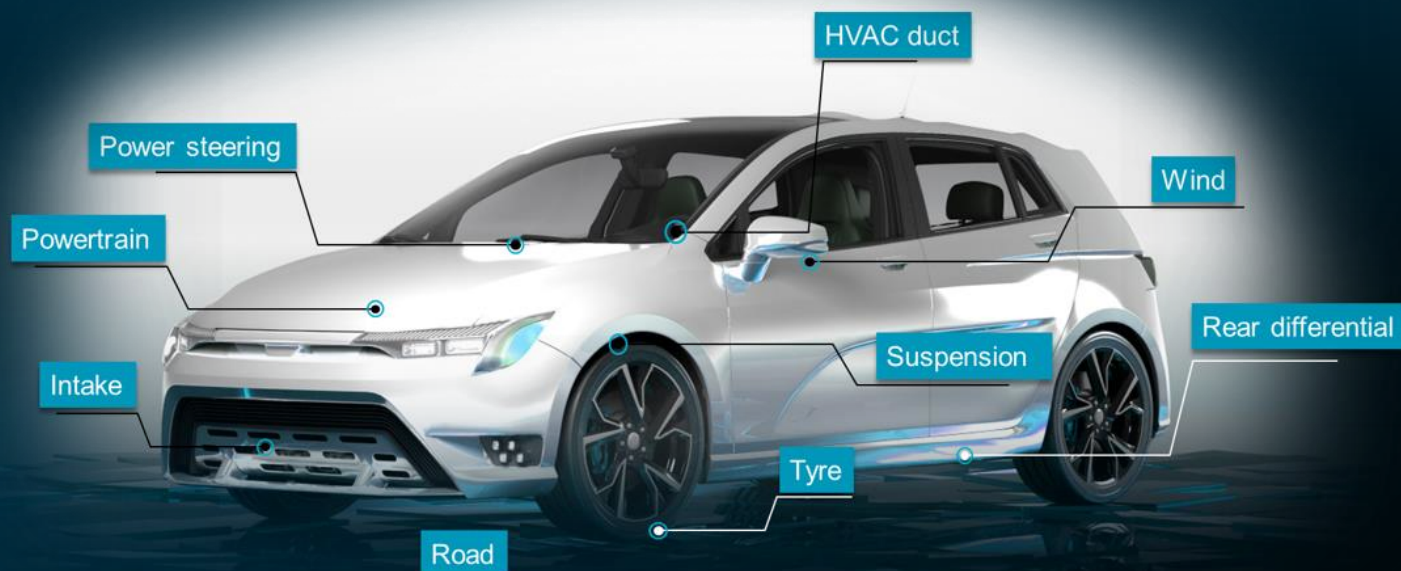
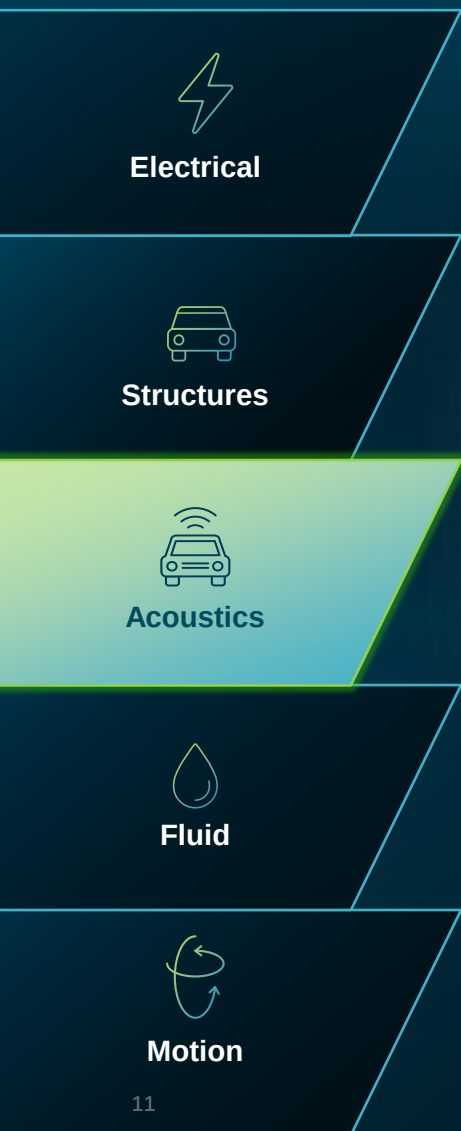


System design

- ✓ 1-D system level provides fully integrated power electronics components with battery pack for simulating a vehicle's driving scenarios



eNVH Requirements



Fluid
Mechanics

ePowertrain &
Drivetrain

Acoustic
Radiation

Structural
Dynamics

Electronics &
Control

Electromagnetic

Target Cascading

Trim development

Pass-by noise

Interior noise

Target Setting

Exterior noise

Drive comfort

Encapsulation

Powertrain NVH

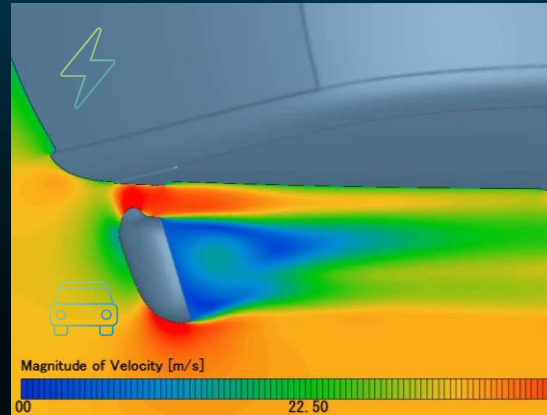
Noise quality

Drivetrain NVH

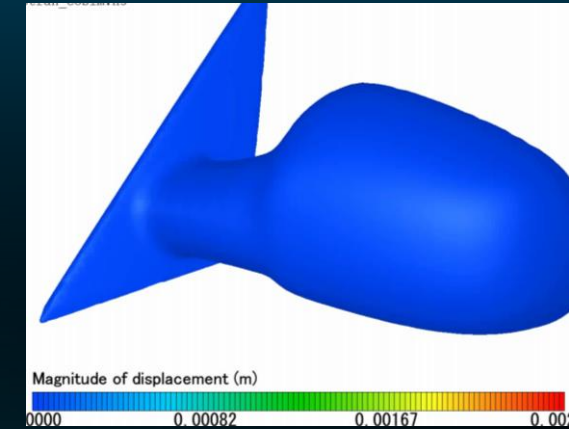
Road/Tyre noise

Co-Simulation | Predicting Operating Condition Performance

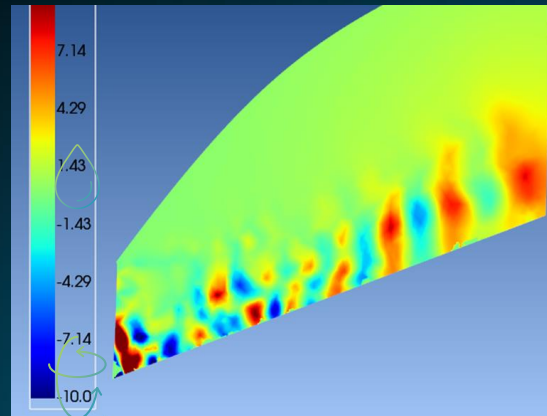
Aero-Vibro-Acoustics workflow



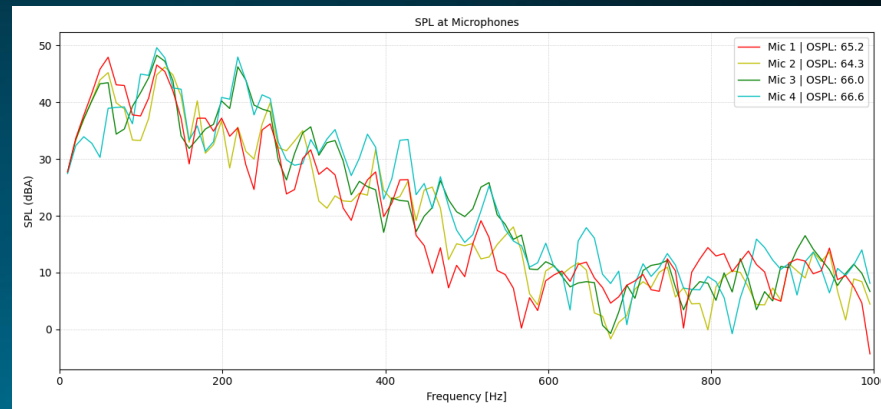
Co-simulation:
1 or 2 way coupled



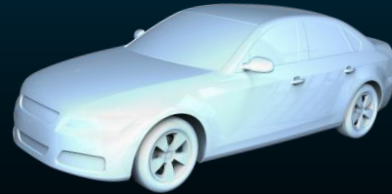
Transient FEA Result



Acoustic & aerodynamic
pressure mapping



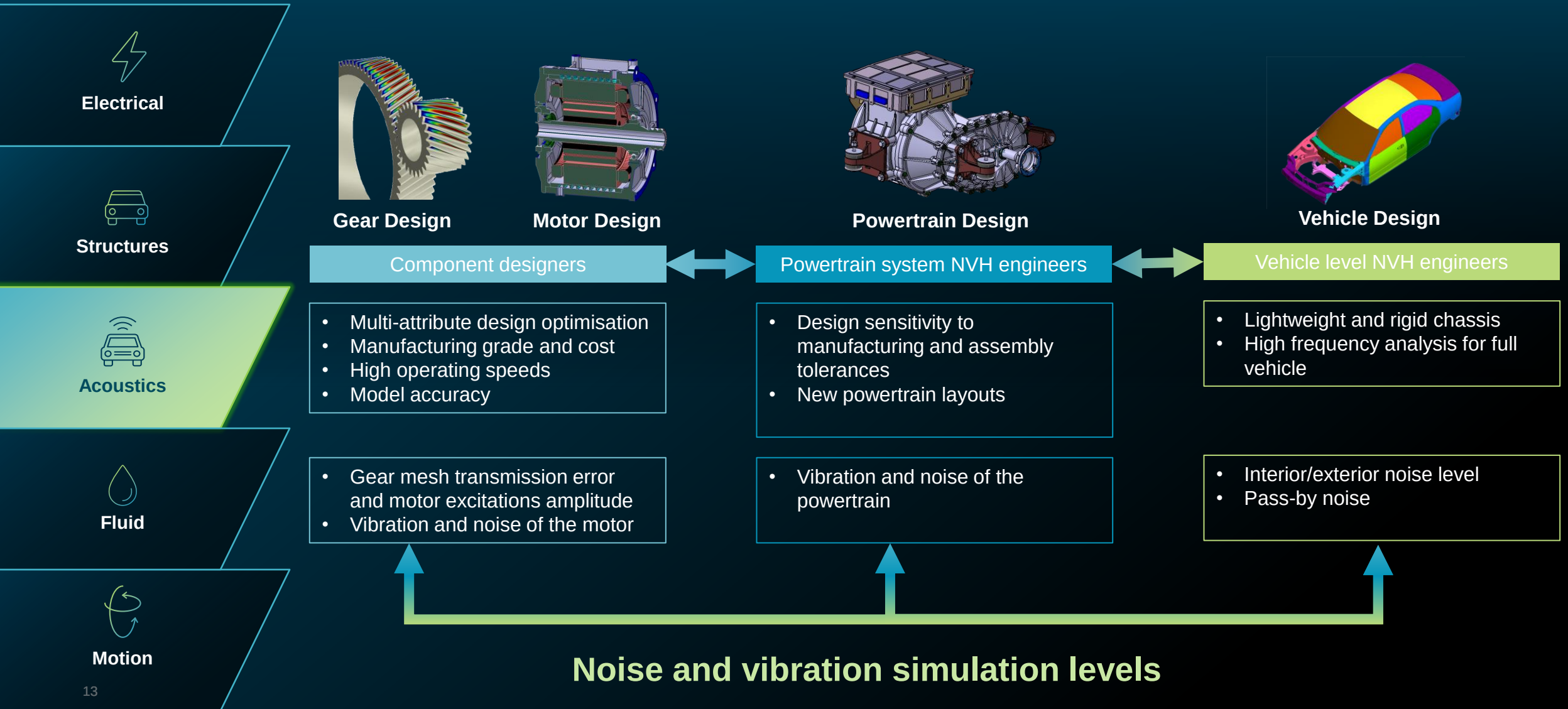
Sound pressure level at interior microphones



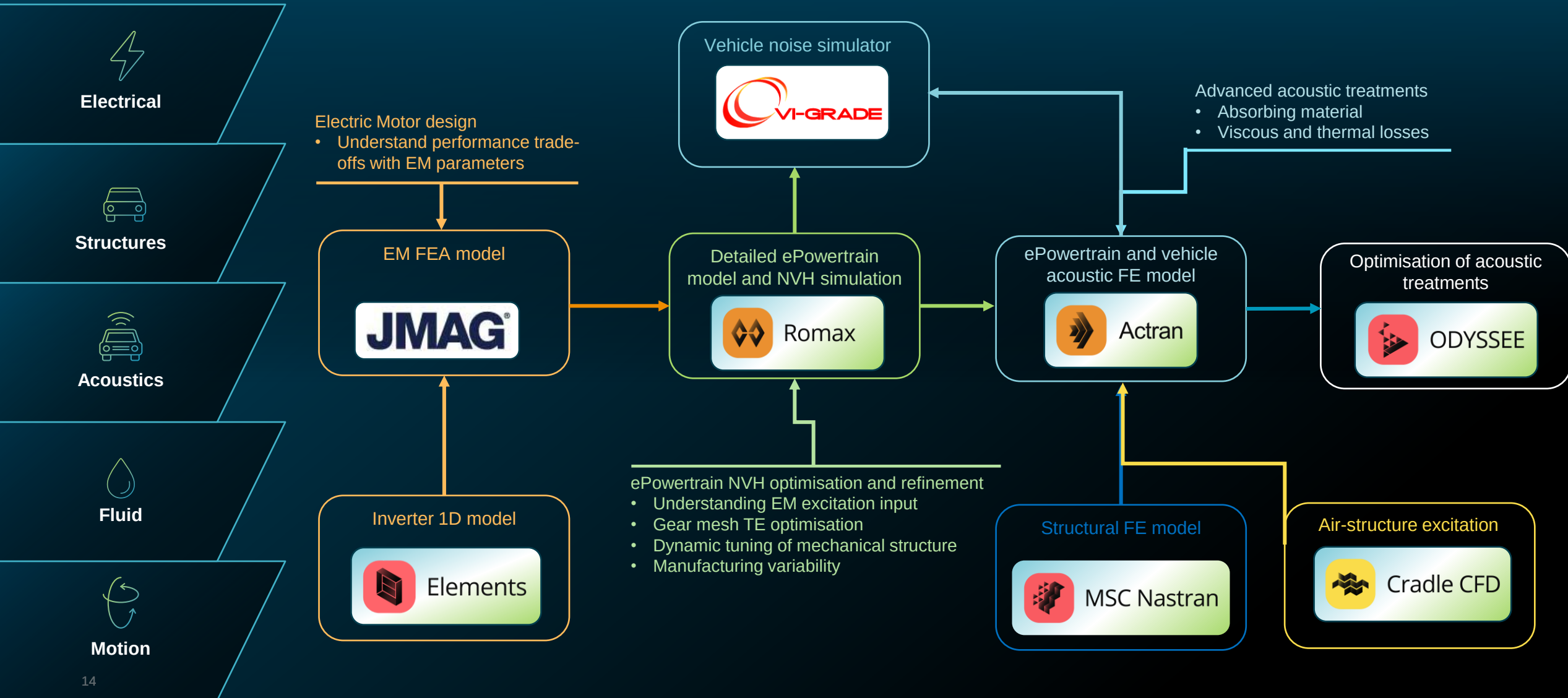
DrivAer Model

- Notchback rear end configuration
- Wind speed: 110 kph

eNVH Requires Cross-functional Collaboration

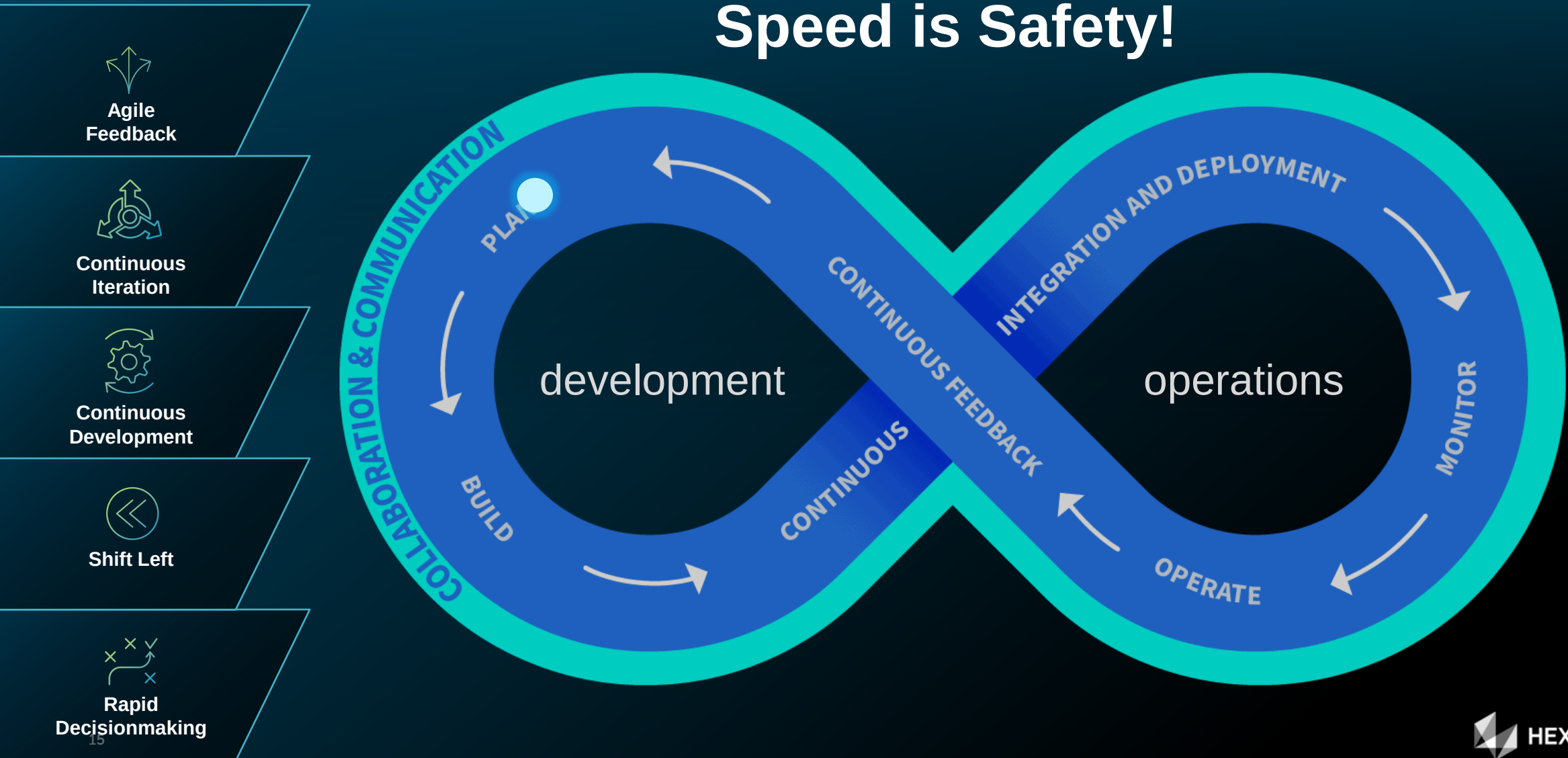


Hexagon's eNVH Solutions are Strengthened by our Partnerships



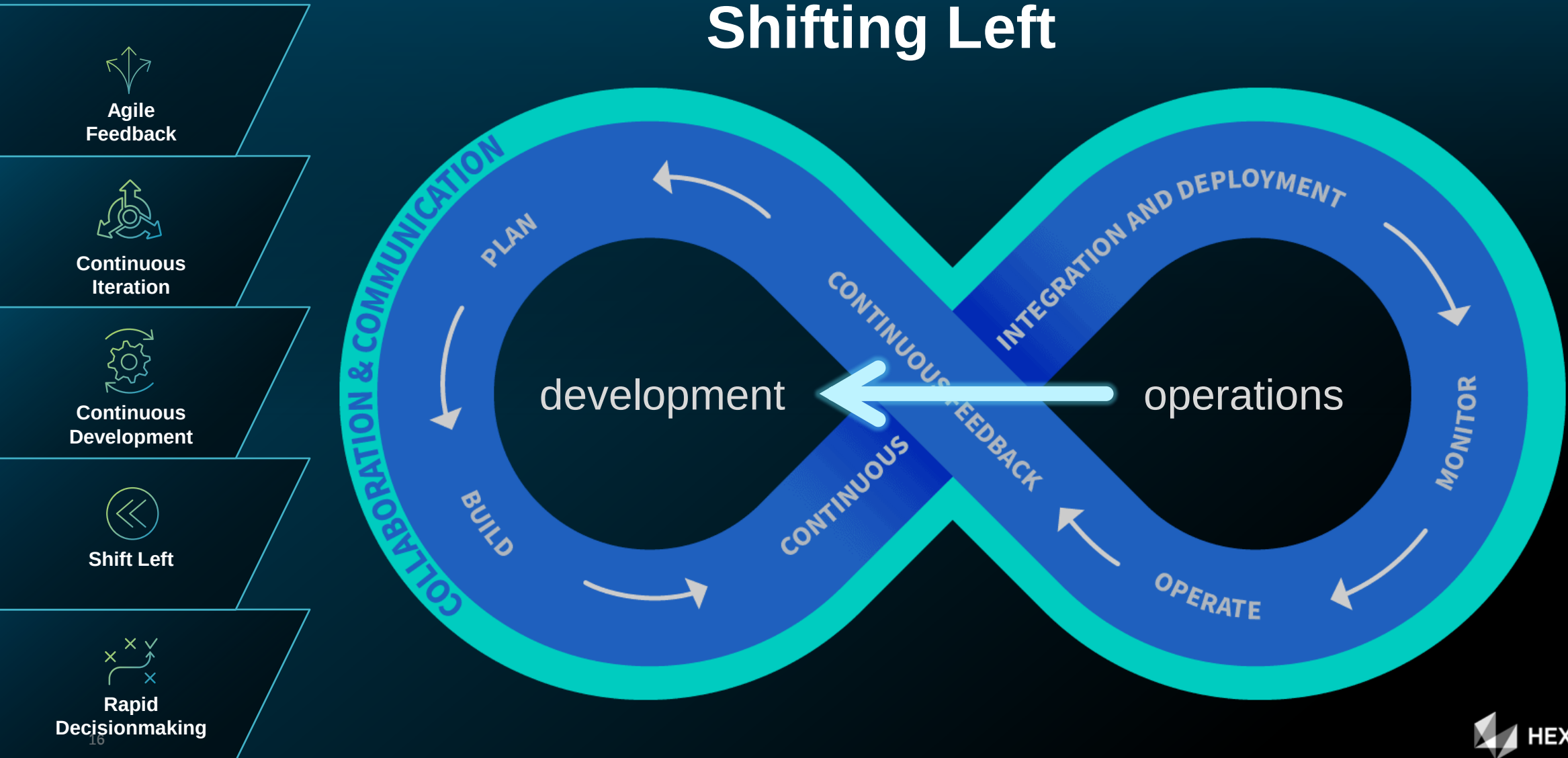
Agile feedback & continuous integration and development

Speed is Safety!

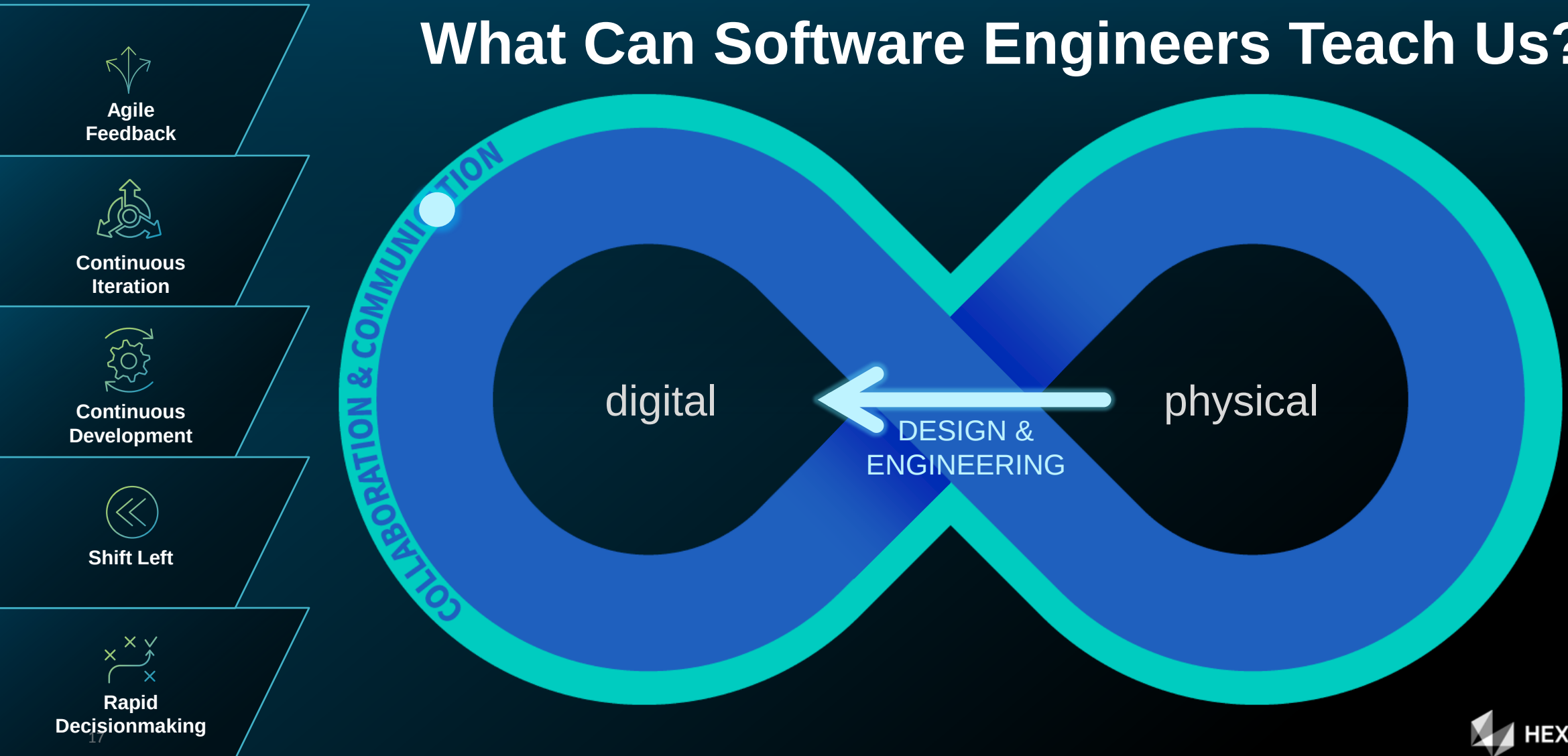


Catching Mistakes and Improving Designs Earlier

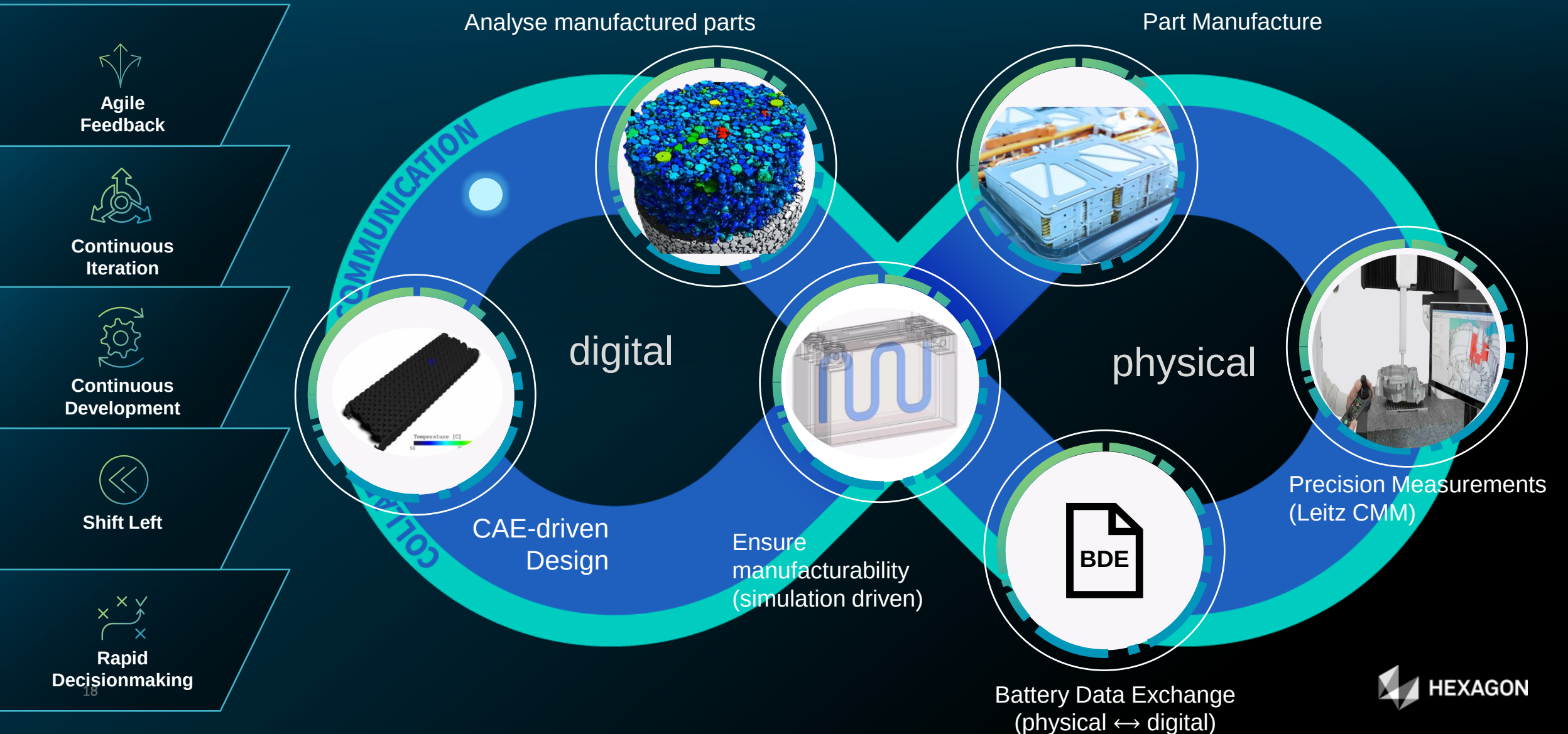
Shifting Left



What Can Software Engineers Teach Us?



Agile Methods| Continuous feedback for battery manufacturing



Elements | A Common Ground across Disciplines for System-Level Solutions


Agile
Feedback


Continuous
Iteration

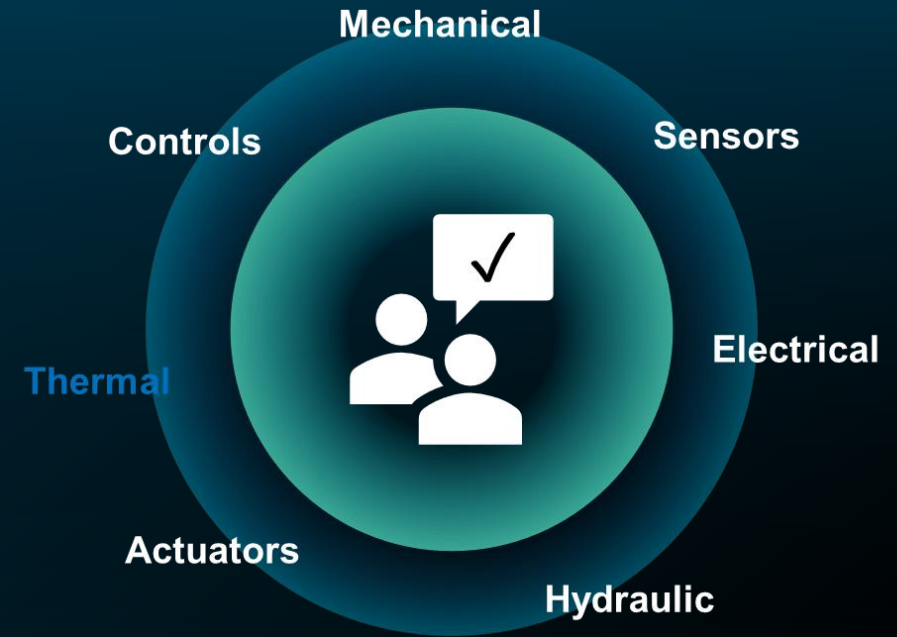

Continuous
Development


Shift Left


Rapid
Decisionmaking



"I want to make this change to improve the thermal performance. But how will it affect the mechanical system?"



Elements

Smart FMU
AI / ML
Smart Superelements
ROM



ODYSSEE

Doing More With Engineering Simulation



**HxGN
Multiphysics**

All physics simulations in one place

Create reliable and accurate digital twins with solutions for structural simulation and analysis, fatigue, computational fluid dynamics, acoustic and NVH simulation and multiphysics co-simulation.



Cradle CFD



Actran



CAEfatigue



Dytran



MSC Nastran



Marc



MSC Apex



Patran



MSC CoSim



**HxGN
System
Dynamics**

Solutions for system-level insight into movement

Enable CAE-led development of complex products with reduced prototyping and enhanced collaboration.



Adams



Adams Modeler



Easy5



Romax Suite



Elements

Powered by **Nexus**



**HxGN
Virtual
Manufacturing**

Solutions for manufacturing process chains

Simulate metal forming, welding and joining, additive manufacturing (AM) and costing.



Simufact Forming



Simufact Welding



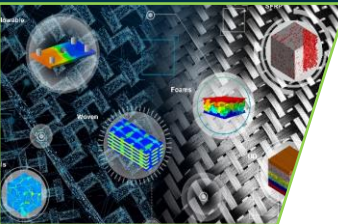
FTI FormingSuite



Simufact Additive



MSC Apex
Generative Design



**HxGN
Digital
Materials**

Solutions for integrated computational material engineering and modelling

Add data management and artificial intelligence capabilities.



Digimat



MaterialCenter



**HxGN
Virtual
Mobility**

Solutions for design, testing, and virtual validation

Ensure safety and quality for ADAS and autonomous vehicle systems._



Virtual Test Drive

Simulation Governance



SimManager



MaterialCenter

Artificial Intelligence & Machine Learning



ODYSSEE

nexus



Thank you.



HEXAGON

empowering an autonomous future

